COMS BC1016

Introduction to Computational Thinking and Data Science

Lecture 15: Estimation and Confidence Intervals

Spring 2025

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November 11, 2026

Project Milestones and Deadlines

NEXT DEADLINE →

1. **Group Declaration**: Deadline **Friday, Nov 6** at 11:59pm
   Please complete the Google Form to indicate your group preference: Final Project Groups. If you do not have a group and would like us to form one for you, there is a space for you to indicate this. Otherwise, you should indicate who you would like to work with. *Only one person needs to submit this per group.*
2. **Project Proposal**: Due **Friday, Nov 20** at 11:59pm
   Each group will select a final project notebook and dataset to work on for the final project and complete the graphs and tables required in the Exploratory Data Analysis section. Groups do not need to include sentence explanations of their graphs and tables in the project proposal. Based on your exploratory data analysis, you will state the hypothesis (null and alternative) you are planning to test.
3. **Progress Report**: Due **Monday, Dec 7** at 11:59pm
   For the progress report, groups should complete the exploratory data analysis section and have mostly completed the hypothesis testing section. Specifically, groups are expected to have a clearly stated null and alternative hypothesis, a clearly stated test statistic, and to have started implementing the hypothesis test. The code for your hypothesis does NOT need to work for the progress report. Your group will only be graded on whether you have started the hypothesis section. Groups should also begin the prediction section. For whatever is not completed in the prediction section by the time of the progress report, groups should list how they plan on approaching it and what analysis they plan to do. Additionally, groups should share if they are running into any issues with their analysis that they may need assistance with or have questions about.
4. **Final Project Report**: Due **Friday, Dec 18** at 11:59pm
   Groups will submit the completed reports along with a completed peer review.

Today's Lecture

- A/B Testing Review
- Percentiles
- Estimation
- Bootstrapping
- Confidence Intervals
- Statistics Summary

Review: A/B Testing

Review: P-Value

P-value: Observed significance level

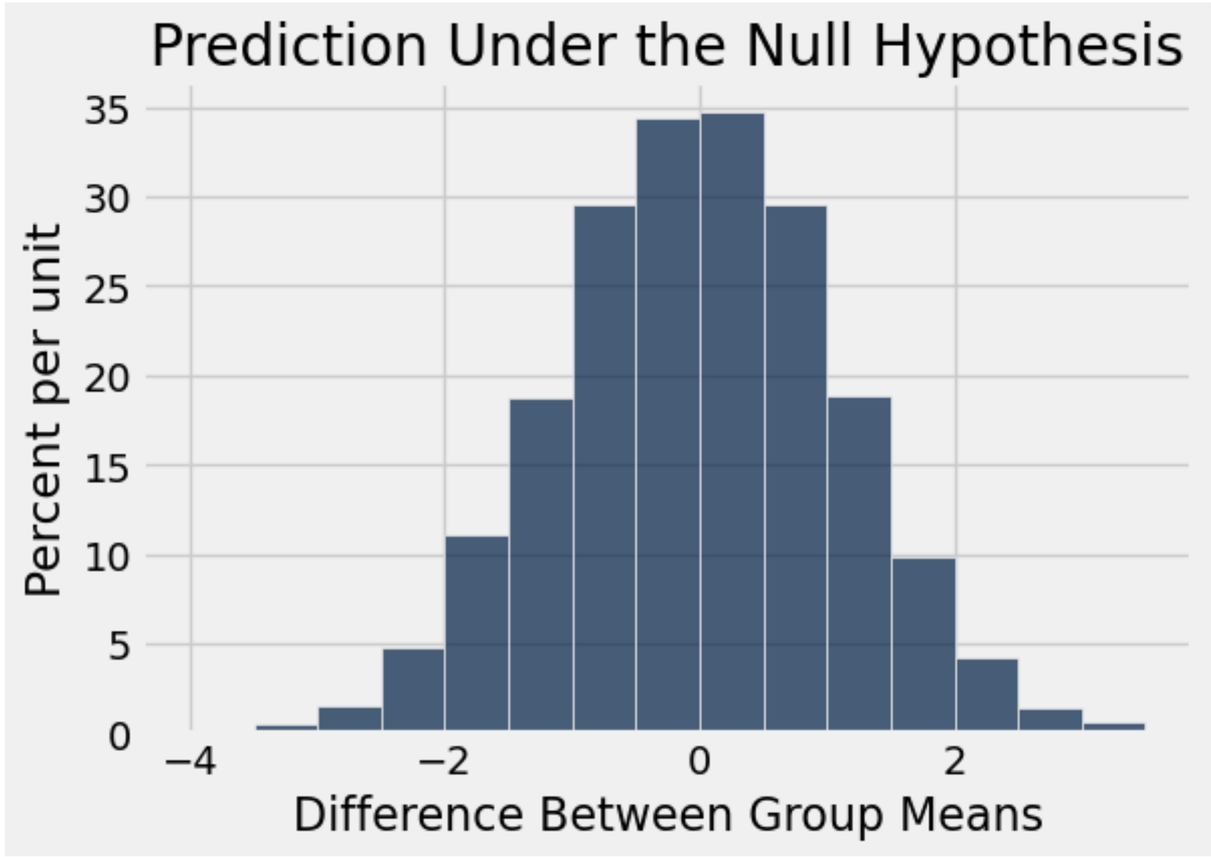
The **P-value** is the chance

under the null hypothesis

that the test statistic

is equal to the value that was observed in the data

or is even further in the direction of the alternative



A/B Testing

- Testing whether Group A and Group B have the same underlying distribution or not
    - Null Hypothesis: The distributions of [test statistic] from both groups are the same
        - Any differences we observe are due to chance
    - Alternative Hypothesis: The distributions are different
- If the distributions look different, it supports the alternative hypothesis

A/B Testing Process

- Under the null, if the Group A and Group B are the same, it doesn't matter what label (A or B) we assign each item
    - We expect the difference between the groups to be 0
- **Permutation Test**:
    - Shuffle all group labels using `shuttle(with_replacement = False)`
    - Compute the difference between the average of the two shuffled groups
    - Compare the difference to our observed difference

Percentiles

Percentile

The p-th percentile of a collection is the smallest value in the collection that is at least as large as $p\%$ of all the values

- Computing the p-th percentile of a list of n values:

    1. Sort the list

    2. Compute how many elements k the $p\%$ refers to

$$k = \left\lceil \frac{p}{100} \times n \right\rceil$$

    3. Return the k-th smallest element (counting from 1)

Percentile

Example: What is the 80th percentile for [12, 17, 6, 9, 7]?

1. Sort the list

2. Compute how many elements k the $p\%$ refers to

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Percentile

Example: What is the 80th percentile for [12, 17, 6, 9, 7]?

1. Sort the list: [6, 7, 9, 12, 17]

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Percentile

Example: What is the 80th percentile for [12, 17, 6, 9, 7]?

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$$k = \left\lceil \frac{p}{100} \times n \right\rceil = \left\lceil \frac{80}{100} \times 5 \right\rceil = \left\lceil \frac{4}{5} \times 5 \right\rceil = 4$$

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   4th smallest element of [6, 7, 9, 12, 17] is 12

`percentile` Function

- `percentile(p, values_array)`
  - Returns the smallest value in `values_array` that is at least as large as `p`% of the elements in the array
  - `p` is always between 0 and 100

Notebook Demo

Example

Let `s = [1, 3, 5, 7, 9]`

What would `percentile(20, s)` evaluate to?

`percentile(p,array)`
returns the smallest value in `array` that is at least as large as `p`% of elements in the array

Example

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More Examples

`percentile(p,array)`
returns the smallest value in `array` that is at least as large as `p`% of elements in the array

Let `s = [1, 3, 5, 7, 9]`

Which of the following evaluate to `True`?

1. `percentile(10, s) == 0`

2. `percentile(39, s) == percentile(40, s)`

3. `percentile(40, s) == percentile(41, s)`

More Examples

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   False! Element is an element from the array

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1. `percentile(10, s) == 0`

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   True! Both evaluate to 3

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More Examples

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   True! Both evaluate to 3

3. `percentile(40, s) == percentile(41, s)`

   False. First value is 3 but the second one is 5

Estimation

Estimation

We use estimation to figure out
the value of an unknown parameter

If entire population is known

Calculate parameter directly

If we only have access to a random sample

We calculate a statistic as an estimate of the parameter

Quantifying Uncertainty

- Our estimate depends on the sample we collected. How different would the estimate be if the sample were different?
    - Can we determine how accurate our estimate is?
    - In theory, we could collect a different sample and check how similar the statistic we calculated is
- What if we can’t go back and collect more samples?

Bootstrap Method

Bootstrap

- A technique for simulating repeated random sampling

- The term “bootstrap” means *to improve or lift yourself using your own resources without external help*

  - From the phrase “to pull oneself up by one’s bootstraps”

- **Bootstrap method** tries to learn about the population using only the sample itself without any outside (theoretical) distribution assumptions

The Bootstrap Method

- Suppose we have a large random sample from the population
  - By the Law of Averages, it probably resembles the population from which it's drawn
  - We can treat it like a miniature version of the population
- We can replicate sampling from the population by sampling from the sample
  - To resample, draw at random with replacement the same number of times as the original sample size

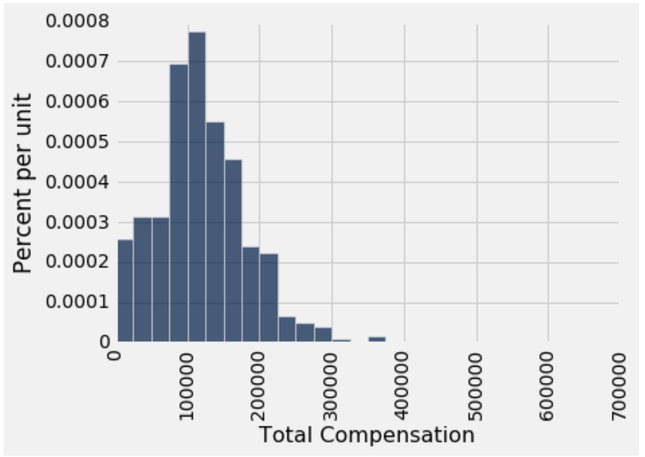
The Bootstrap

Population

?

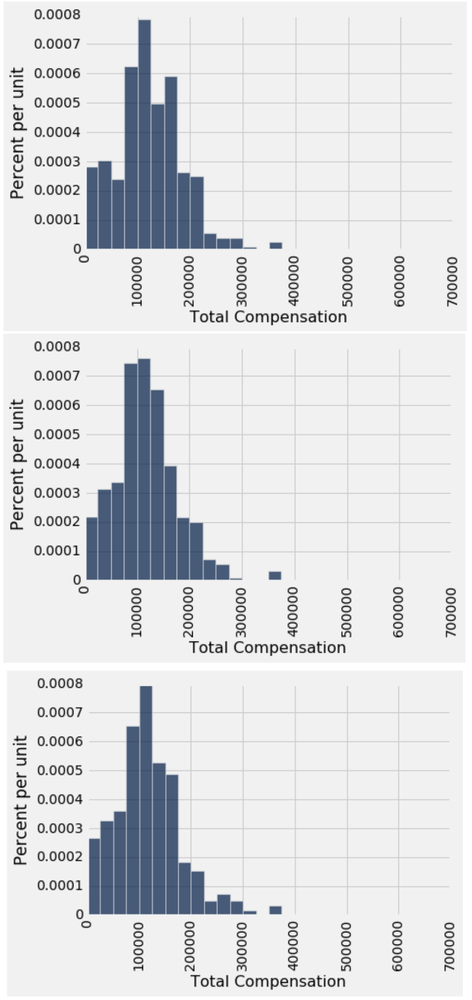
We don’t know the entire population and thus can’t calculate the parameter directly

Sample



However, we can take a single sample…

Resamples



…and generate lots of resamples

Notebook Demo: SF Gov’t Salaries

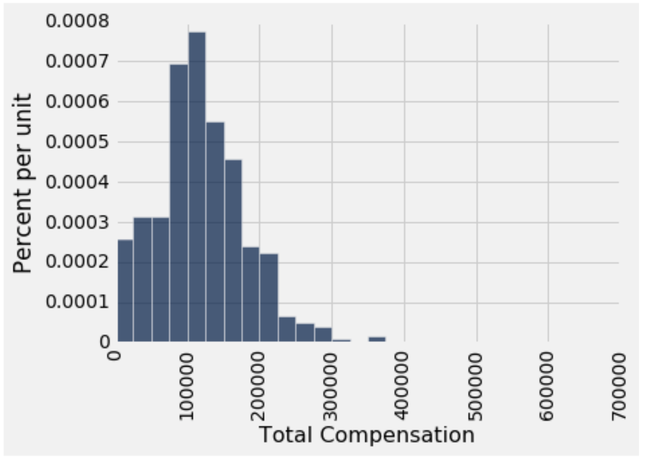
The Bootstrap

Population

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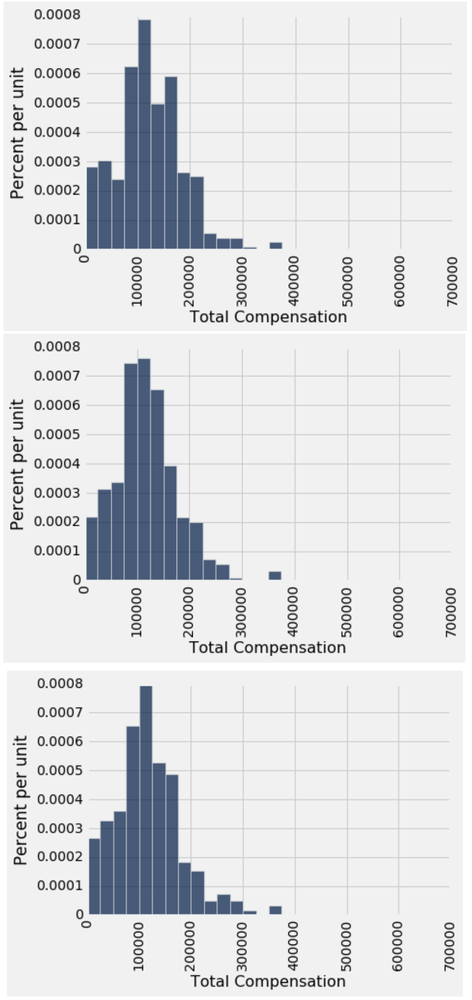
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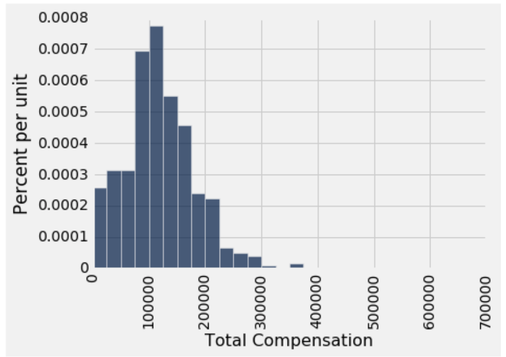
Resamples



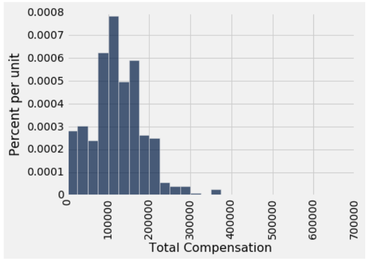
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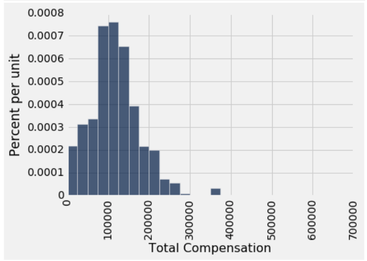
Confidence Interval

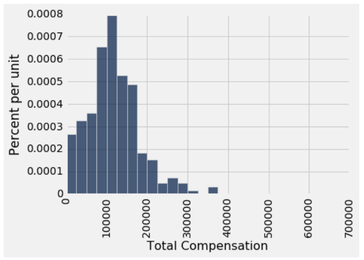
Sample



Resamples







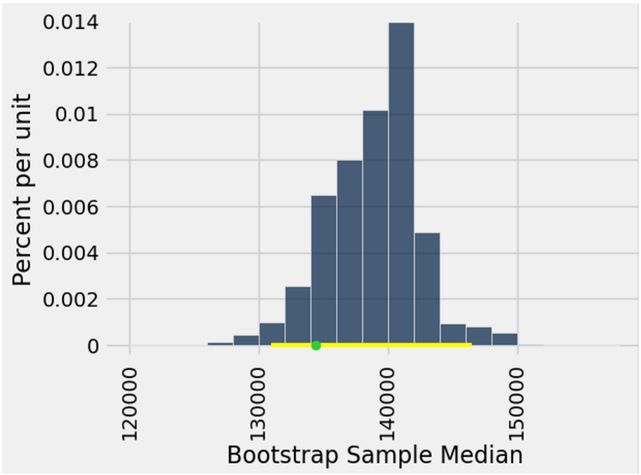
Statistic

median_1

median_2

median_3

Distribution of the Statistic



Confidence Intervals

Confidence Interval

- Interval of **estimates of a parameter**
  - How confident we are that the parameter (the real value calculated from the population) is likely to be within this interval
    - “How often does our estimate capture the parameter”
- Based on random sampling
  - 95% confidence interval is the middle 95% of the distribution of estimates
    - Can choose any percentage between 0 and 100
    - Higher numbers correspond to a wider interval

Interpreting Confidence Interval

- The confidence interval helps us state how confident we are that our sample statistic estimates (or contains) the true population parameter

- When we do bootstrapping and create, say, a 95% confidence interval for the median, what we are doing is estimating the range of plausible values for the true population median based on the sample data

- Confidence interval is a rough estimate of how large a parameter is. Common mistake to use it for other purposes

    - For example, a 95% confidence interval for an average age is **not** saying we think 95% of the population is within that age

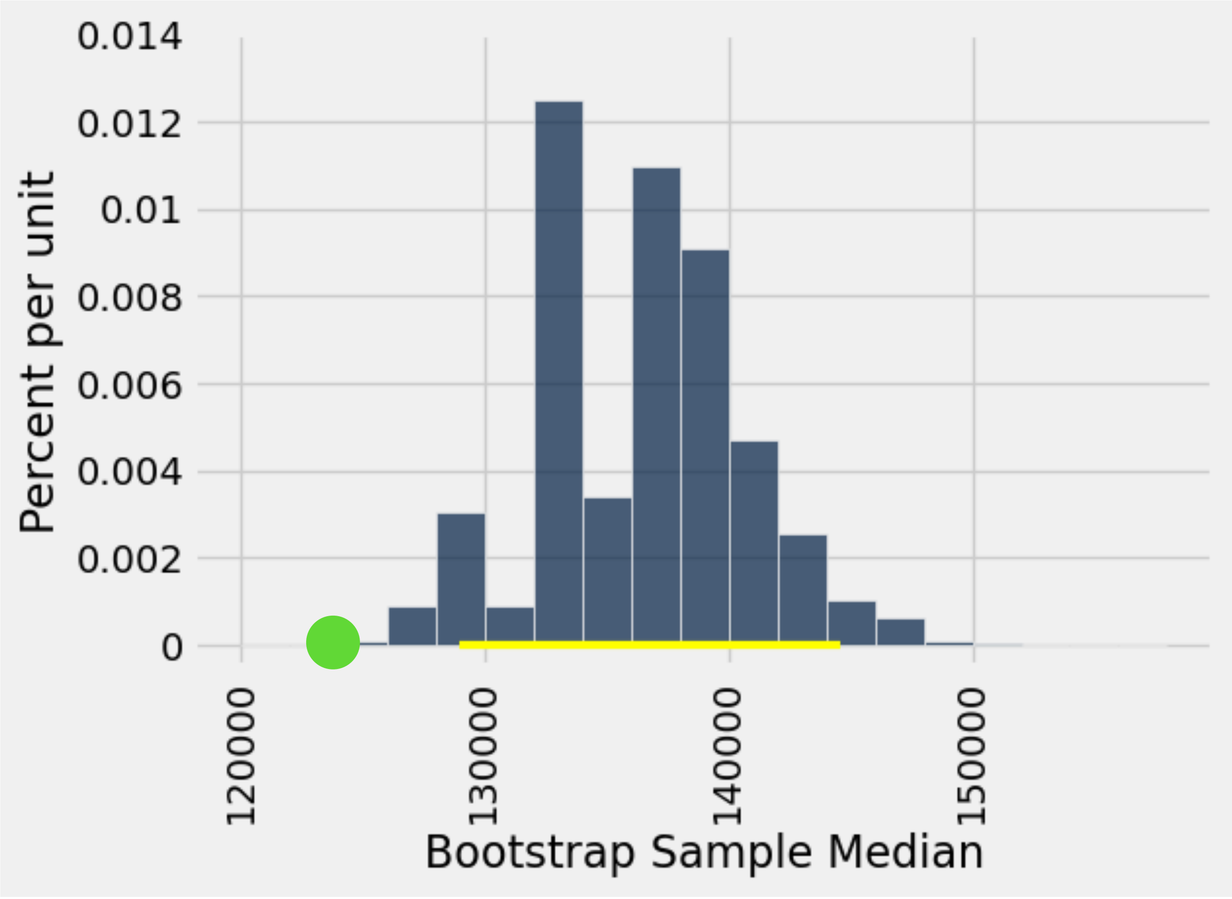
Relationship between Confidence Interval and P-value

- For a p-value cutoff of $p\%$, we reject the null hypothesis if the value is not within $(100 - p)\%$ confidence interval
  - If you use a $p\%$ cutoff for the p-value and the null hypothesis is true, then there is about a $p\%$ chance your test will conclude the alternative is true
    - If the null hypothesis is true, $(100 - p)\%$ chance test agrees with the null
  - $(100 - p)\%$ confidence interval says we're $(100 - p)\%$ certain the parameter is somewhere within the interval
    - If the value is outside of the interval, we reject the null hypothesis

Relationship between Confidence Interval and P-value

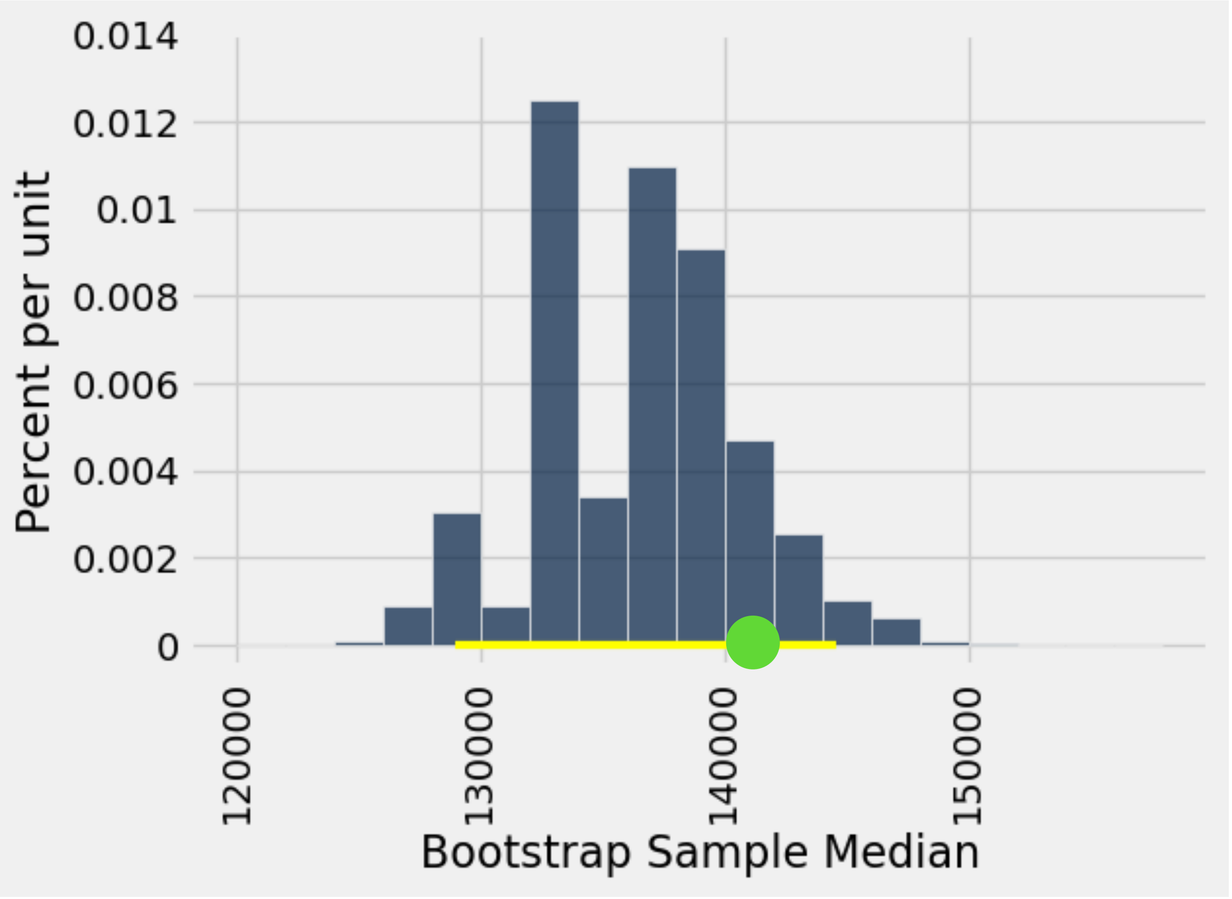
- Example:
  - Null hypothesis: Population average $= x$
  - Alternative Hypothesis: Population average $\neq x$
  - If x is not in our $(100 - p)\%$ interval, then we reject the null

Rejecting the null



Our x is outside the 95% confidence interval

Cannot reject null



Our x is inside the 95% confidence interval

Confidence Interval & P-Value Example

Null Hypothesis: You have a fair coin with 50% probability of getting heads or tails

Alternative: The coin is biased

Your observed value for **% heads** is 65%

Let's say your 95% confidence interval is [45, 60]

Confidence Interval & P-Value Example

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Questions:

1. For a 5% p-value cutoff, can we reject the null?

2. For a 10% p-value cutoff, can we reject the null?

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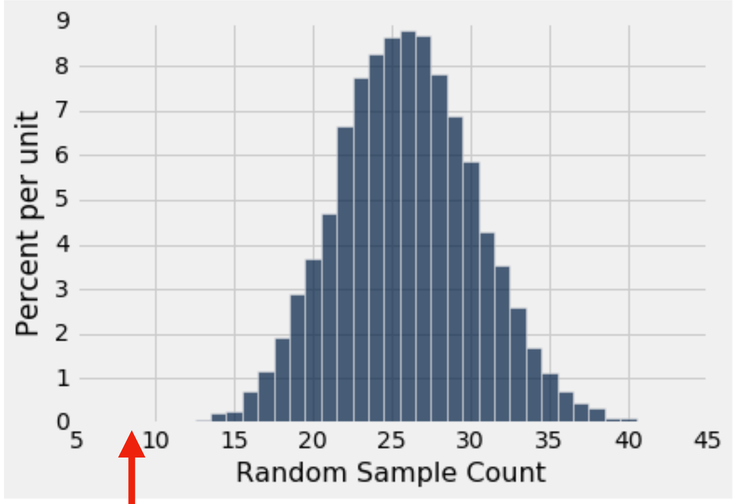
1. For a 5% p-value cutoff, can we reject the null?
   - Yes, 65% is outside our confidence interval
2. For a 10% p-value cutoff, can we reject the null?
   - Yes, we expect the confidence interval to be even narrower, so 65% would still be outside the confidence interval

Summary of Stats so far…

Hypothesis Testing

- Modeling expected outcomes under the null and comparing it to our observed outcome

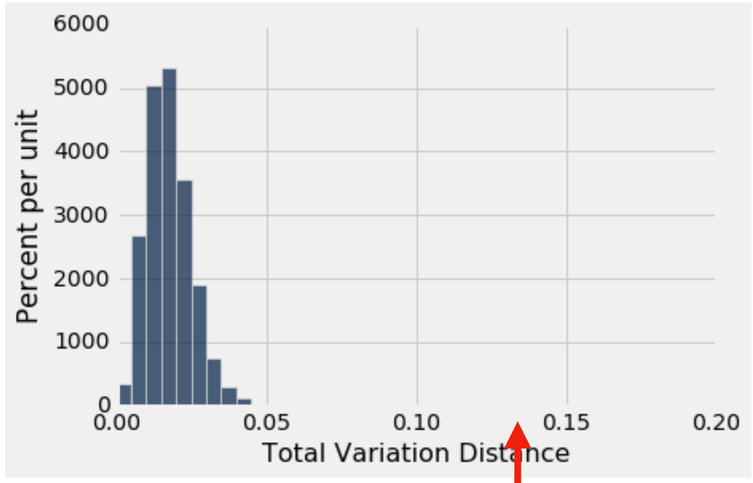
Swain v Alabama



Observed Number

2 Categories

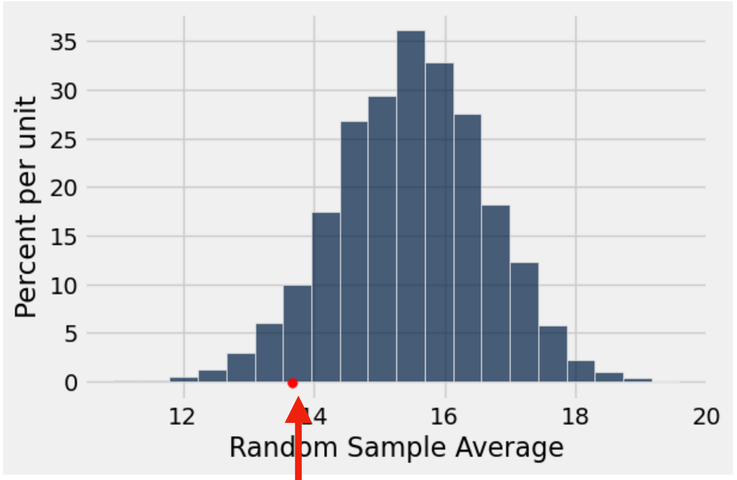
Alameda Jury



Observed TVD

3+ Categories

Midterm Exam Scores



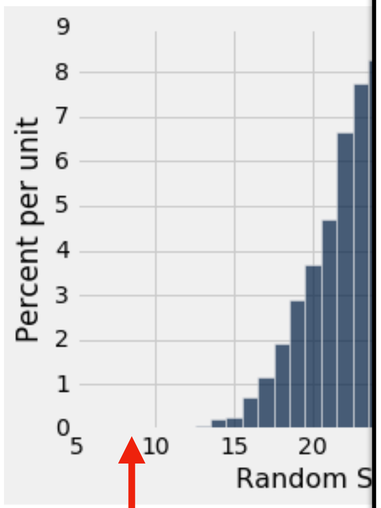
Observed Average

Numerical Data

Hypothesis Testing

- Modeling expected outcomes under the null and comparing it to our observed outcome

Swain v



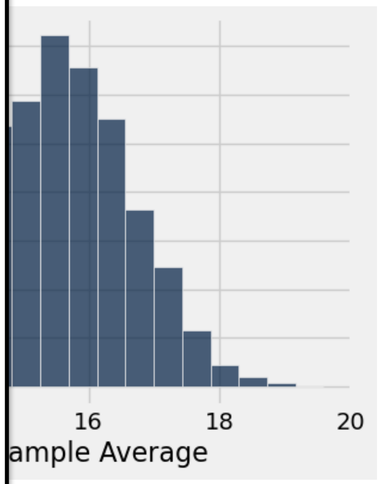
Observed N

It's often not easy to say whether the observed outcome falls within our expectations…

How can we more precisely characterize the likelihood of observing an expected outcome?

p-value!

kam Scores



Average

2 Categories

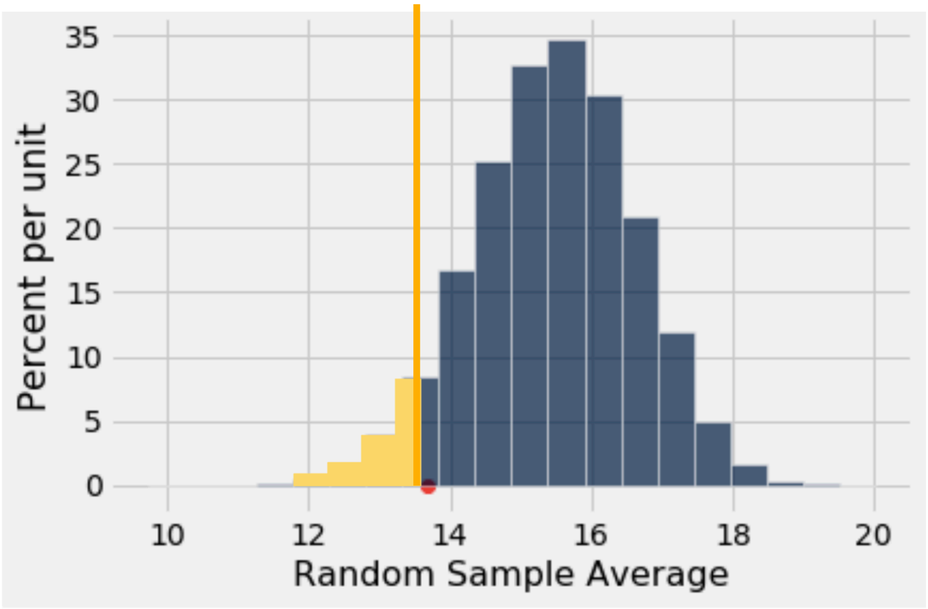
3+ Categories

Numerical Data

Hypothesis Testing

- Modeling expected outcomes under the null and comparing it to our observed outcome

Midterm Exam Scores



p-value = 0.058

p-value & **statistical significance**

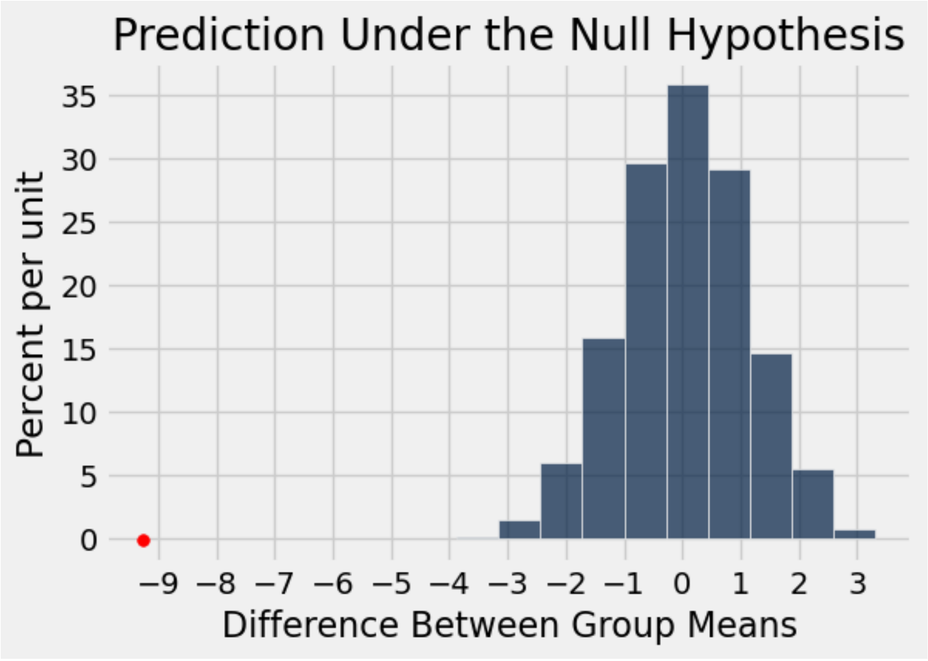
Process:

- Calculate the area of the tail (to the left/right of our observed value)

Hypothesis Testing

- Modeling expected outcomes under the null and comparing it to our observed outcome

Smoking vs Non-Smoking Mothers & Birthweight



p-value = 0

A/B Testing

Compare the difference between two groups

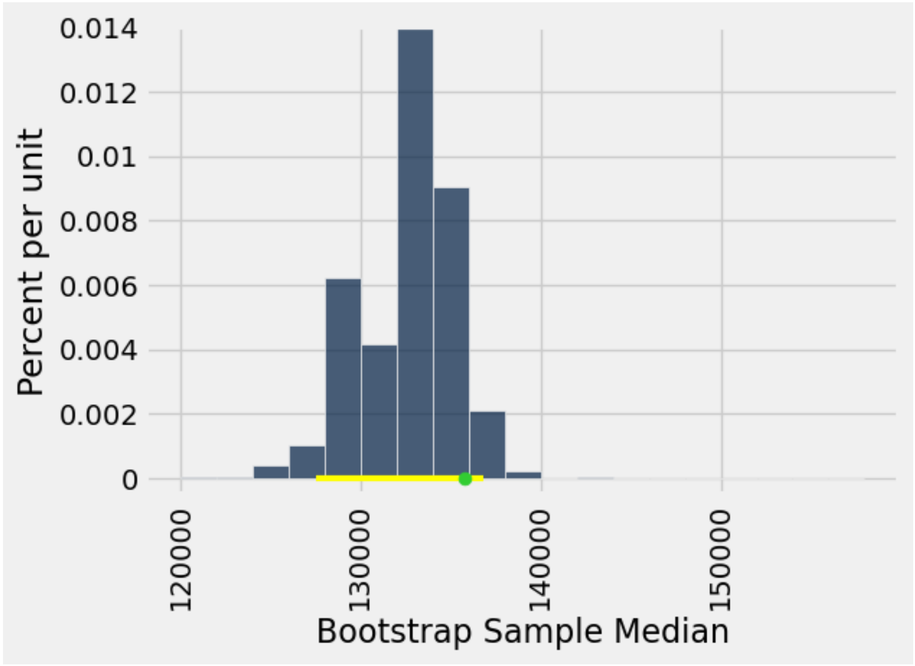
Process:

- Permutation test (shuffle labels)

Estimating a Parameter

- We want to estimate a population parameter from a sample statistic

Median Employee Salary



95% Confidence Interval: Median Salary between $125,745 and $140,318

Confidence Interval

Lets us estimate a range for what we think the parameter's value is

Process:

- Bootstrap

Next time

- Normal Distributions